

## Performance Testing

|                      |                                   |
|----------------------|-----------------------------------|
| <b>Date</b>          | 02 July 2026                      |
| <b>Team ID</b>       |                                   |
| <b>Project Name</b>  | Personalized Networking Assistant |
| <b>Maximum Marks</b> | 5 Marks                           |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                                                     |
|----------------------------|-----------------------------------------------------------------------------|
| <b>Testing Tool Used</b>   | Pytest, FastAPI TestClient (httpx), Manual UI Testing                       |
| <b>Type of Testing</b>     | Unit Testing, Functional Testing, API Testing                               |
| <b>Target Module / API</b> | Event Analyzer, Topic Generator, Fact Checker, FastAPI Routes               |
| <b>Test Environment</b>    | Local Development Environment (Windows OS, Python 3.12, FastAPI, Streamlit) |
| <b>Test Date</b>           | 03 July 2026                                                                |

#### Step 2: Test Scenarios

| S. No | Test Scenario / Description                             | No. of Virtual Users | Duration (sec) | Expected Outcome                                               |
|-------|---------------------------------------------------------|----------------------|----------------|----------------------------------------------------------------|
| 1     | Test Event Analyzer using sample event descriptions     | 1                    | 2 – 3          | Event themes extracted successfully using DistilBERT           |
| 2     | Test Topic Generator through Streamlit UI               | 1                    | 2 – 4          | Personalized conversation starters generated successfully      |
| 3     | Test Fact Checker module using Wikipedia API            | 1                    | 1 – 3          | Relevant fact summaries retrieved successfully                 |
| 4     | Test FastAPI routes using Pytest and FastAPI TestClient | 1                    | 2              | API endpoints responded with expected status codes and outputs |

### Step 3: Performance Test Results

| S.No | Metric              | Target Value | Actual Value                                | Status (Pass / Fail) | Remarks                                          |
|------|---------------------|--------------|---------------------------------------------|----------------------|--------------------------------------------------|
| 1    | Response Time (Avg) | < 4 seconds  | Within expected range                       | Pass                 | Observed during manual testing                   |
| 2    | Response Time (Max) | < 8 seconds  | Within expected range                       | Pass                 | No significant delays observed                   |
| 3    | API Availability    | 100%         | All tested endpoints responded successfully | Pass                 | Verified using FastAPI TestClient and Swagger UI |
| 4    | Error Rate          | 0%           | No errors observed                          | Pass                 | During manual and API testing                    |
| 5    | CPU Utilization     | N/A          | Not Measured                                | N/A                  | Resource monitoring not performed                |
| 6    | Memory Utilization  | N/A          | Not Measured                                | N/A                  | Resource monitoring not performed                |

### Step 4: Observations & Analysis

#### Key Findings:

- Event Analyzer successfully extracted relevant themes from networking event descriptions.
- GPT-2 generated meaningful and context-aware conversation starters.
- Wikipedia API returned concise summaries for fact verification.
- FastAPI endpoints functioned correctly during API testing.
- Streamlit frontend provided a smooth end-to-end user experience.
- All implemented pytest test cases executed successfully.

#### Bottlenecks Identified:

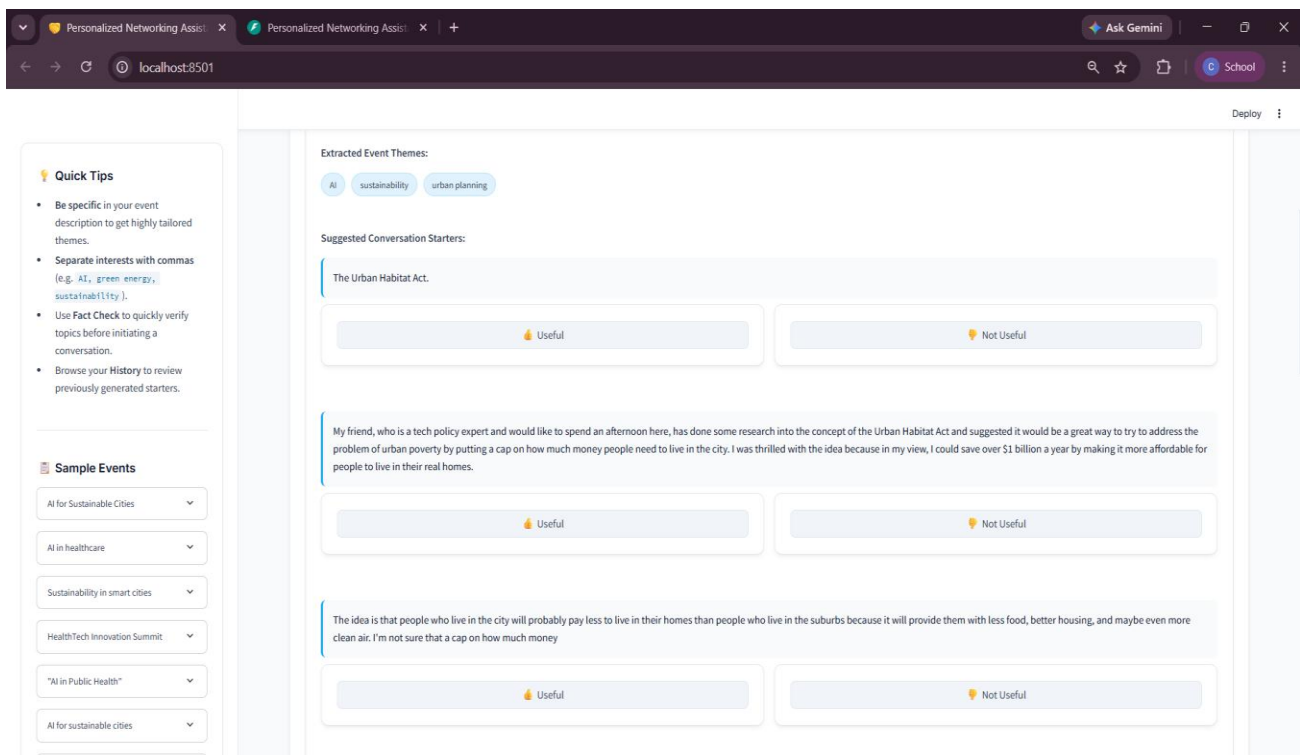
- Initial application startup takes longer because DistilBERT and GPT-2 models are downloaded and loaded into memory on the first run.
- Wikipedia fact verification depends on internet connectivity and API response time.
- GPT-2 generation time increases slightly for longer event descriptions.

#### Optimization Steps Taken:

- Used DistilBERT, a lightweight transformer model, for efficient theme extraction.
- Cached downloaded Hugging Face models locally for faster subsequent executions.
- Implemented modular architecture separating frontend, backend, and AI services.
- Added input validation to reduce invalid API requests.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Streamlit application displaying generated conversation starters

```

• (venv) PS C:\SEM5\Internship_Smartbridge\Project> pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.11.7, pytest-8.3.5, pluggy-1.6.0 -- C:\SEM5\Internship_Smartbridge\Project\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\SEM5\Internship_Smartbridge\Project
plugins: anyio-4.14.1
collected 27 items

tests/test_event_analyzer.py::test_event_analysis_returns_list PASSED [ 3%]
tests/test_event_analyzer.py::test_event_analysis_returns_at_most_three PASSED [ 7%]
tests/test_event_analyzer.py::test_event_analysis_returns_non_empty PASSED [ 11%]
tests/test_event_analyzer.py::test_event_analysis_labels_from_candidates PASSED [ 14%]
tests/test_event_analyzer.py::test_event_analysis_custom_labels PASSED [ 18%]
tests/test_event_analyzer.py::test_event_analysis_short_description PASSED [ 22%]
tests/test_fact_checker.py::test_fact_checker_returns_summary PASSED [ 25%]
tests/test_fact_checker.py::test_fact_checker_returns_correct_extract PASSED [ 29%]
tests/test_fact_checker.py::test_fact_checker_missing_extract PASSED [ 33%]
tests/test_fact_checker.py::test_fact_checker_handles_connection_error PASSED [ 37%]
tests/test_fact_checker.py::test_fact_checker_handles_timeout PASSED [ 40%]
tests/test_fact_checker.py::test_fact_checker_handles_generic_exception PASSED [ 44%]
tests/test_routes.py::test_root_health_check PASSED [ 48%]
tests/test_routes.py::test_generate_conversation_api PASSED [ 51%]
tests/test_routes.py::test_generate_conversation_returns_non_empty_suggestions PASSED [ 55%]
tests/test_routes.py::test_generate_conversation_invalid_request_422 PASSED [ 59%]
tests/test_routes.py::test_generate_conversation_missing_interests_422 PASSED [ 62%]
tests/test_routes.py::test_analyze_event_api PASSED [ 66%]
tests/test_routes.py::test_analyze_event_invalid_422 PASSED [ 70%]
tests/test_routes.py::test_fact_check_api PASSED [ 74%]
tests/test_routes.py::test_fact_check_invalid_422 PASSED [ 77%]
tests/test_topic_generator.py::test_topic_generation_returns_list PASSED [ 81%]
tests/test_topic_generator.py::test_topic_generation_returns_non_empty PASSED [ 85%]
tests/test_topic_generator.py::test_topic_generation_returns_strings PASSED [ 88%]
tests/test_topic_generator.py::test_topic_generation_non_empty_strings PASSED [ 92%]
tests/test_topic_generator.py::test_topic_generation_at_most_three PASSED [ 96%]
tests/test_topic_generator.py::test_topic_generation_empty_interests PASSED [100%]

27 passed in 196.87s (0:03:16)

```

FastAPI Swagger UI showing successful API execution

```

• (venv) PS C:\SEM5\Internship_Smartbridge\Project> pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.11.7, pytest-8.3.5, pluggy-1.6.0 -- C:\SEM5\Internship_Smartbridge\Project\venv\Scripts\python.exe
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tests/test_topic_generator.py::test_topic_generation_empty_interests PASSED [100%]

===== 27 passed in 196.87s (0:03:16) =====

```

Terminal output showing successful execution of pytest